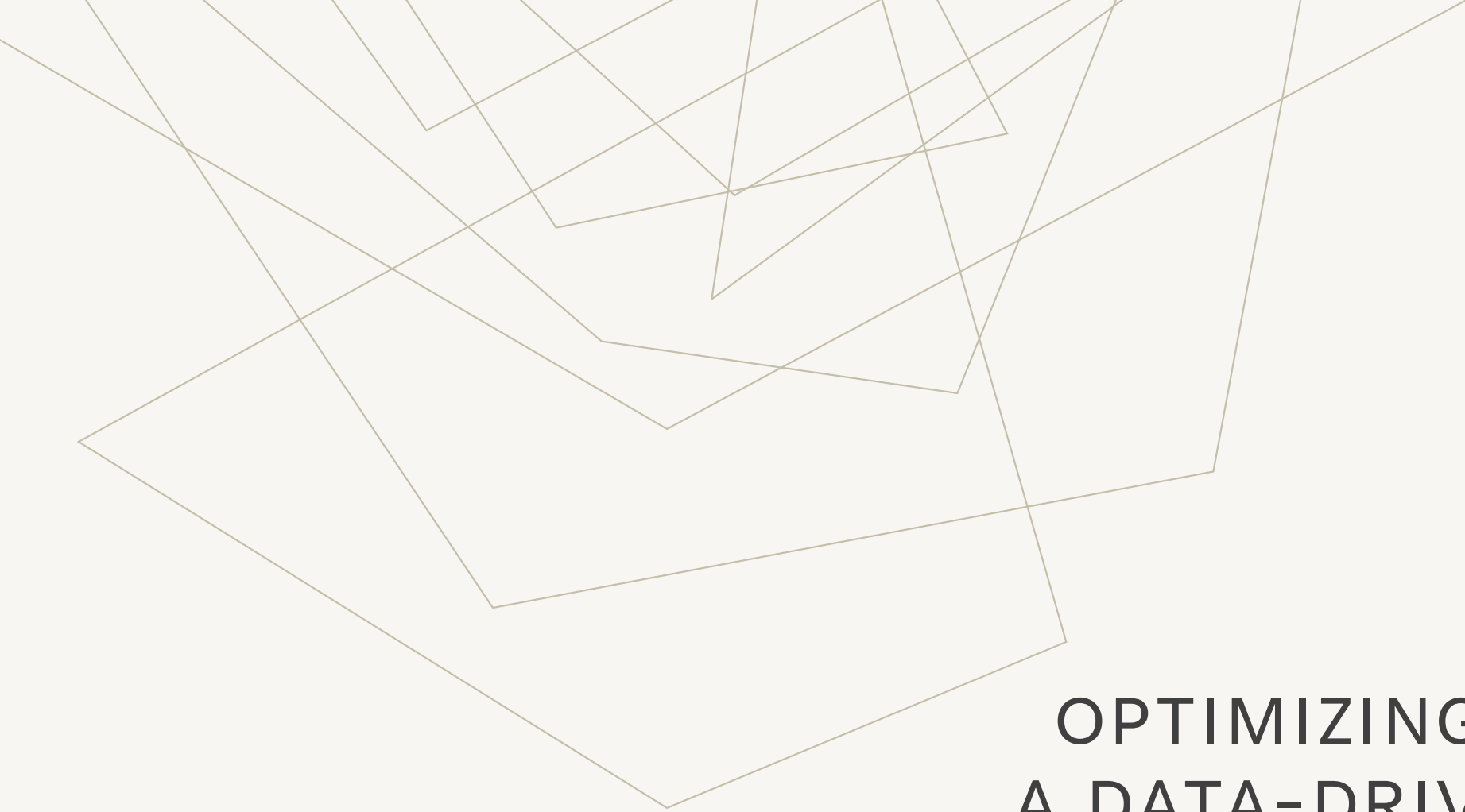


Abstract geometric lines in the top left corner, consisting of several thin, light brown lines that intersect to form various polygons and shapes, creating a modern, architectural feel.

OPTIMIZING AIR TRAVEL: A DATA-DRIVEN APPROACH TO FLIGHT DELAY ANALYSIS AND PREDICTION.

VIKRANT SINGH [22112114]

DATA CLEANING & EXPLORATION

- **Dropped 0.50% missing values** in the age column assuming they were **Missing at Random (MAR)** based on stable distribution before and after removal.
- **No duplicate rows** were found in the dataset.
- **Density plot of age** remained consistent before and after dropping missing values, supporting the MAR assumption.

FEATURE ENG. & CONSTRUCTION

arr_flights – Arrival flights
month_sin – Month sine
month_cos – Month cosine
carrier_ct – Carrier delay
weather_ct – Weather delay
nas_ct – NAS delay
security_ct – Security delay
late_aircraft_ct – Aircraft delay
carrier – Airline code
airport – Airport code

TARGET LABEL CONSTRUCTION FOR CLASSIFICATION

```
df = df[df["arr_flights"] > 0]
df["arr_del15_rate"] = df["arr_del15"] / df["arr_flights"]
df["is_delayed"] = (df["arr_del15_rate"] > 0.05).astype(int)
```

Identifies airports with a delay rate over 5%, creating a binary target for delay prediction.

ROOT CAUSE ANALYSIS

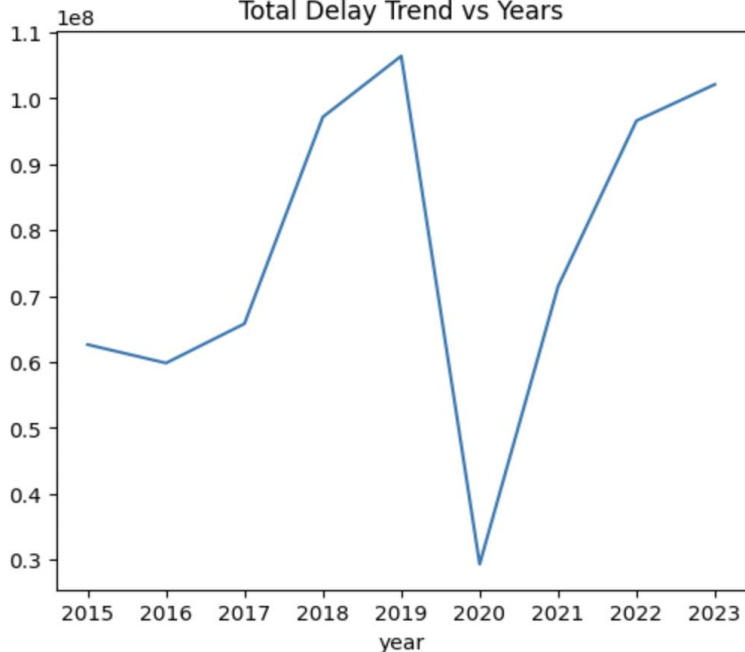
- **Carrier Delay** – Crew scheduling, maintenance
- **NAS Delay** – Air traffic control, congestion
- **Weather Delay** – Storms, low visibility
- **Late Aircraft** – Previous flight delays
- **Security Delay** – Airport security checks

IDENTIFIED BOTTLENECKS

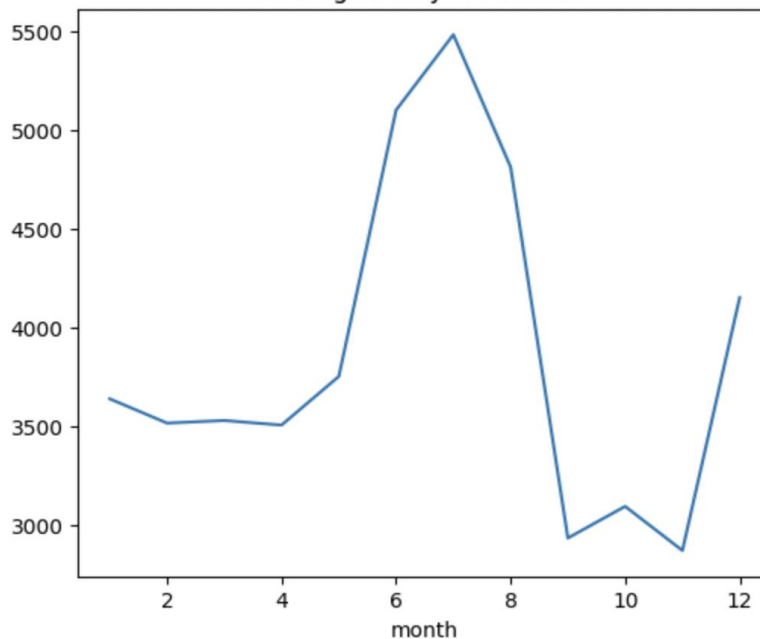
- High delay frequency at major hubs (e.g., ATL, ORD).
- Seasonal weather delays spike in winter and summer.
- Late aircraft is a compounding delay factor.

TRENDS

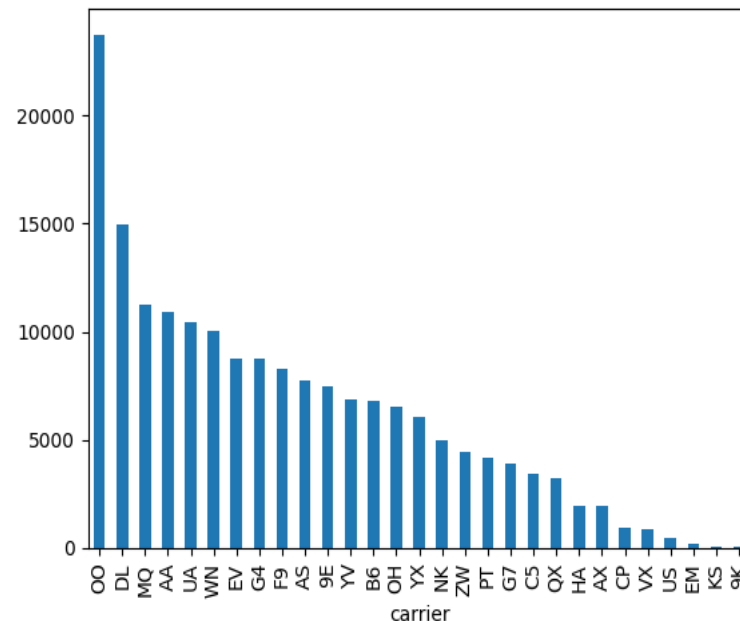
Total Delay Trend vs Years



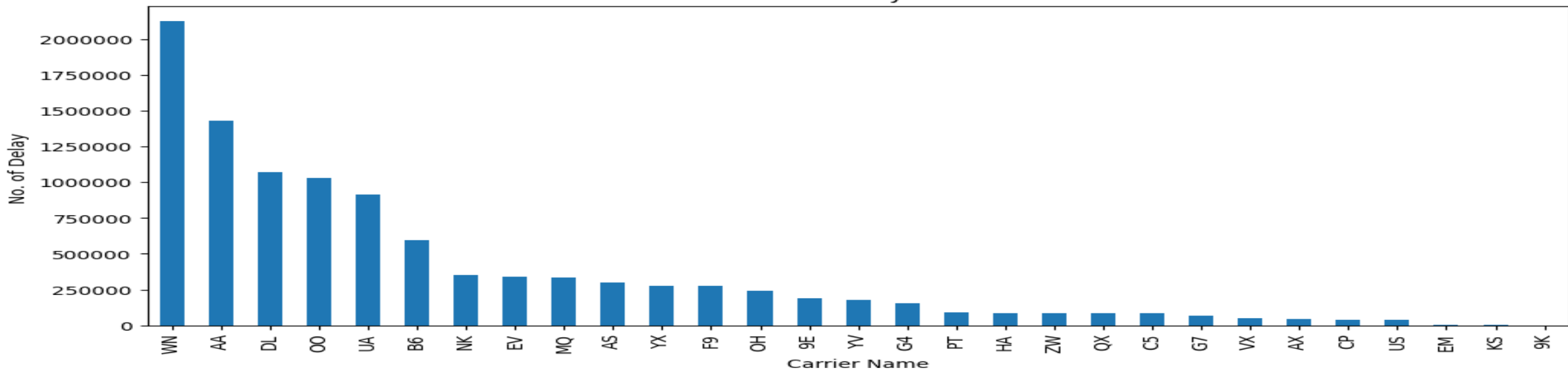
Average Delay vs Months



Value Count vs Carrier

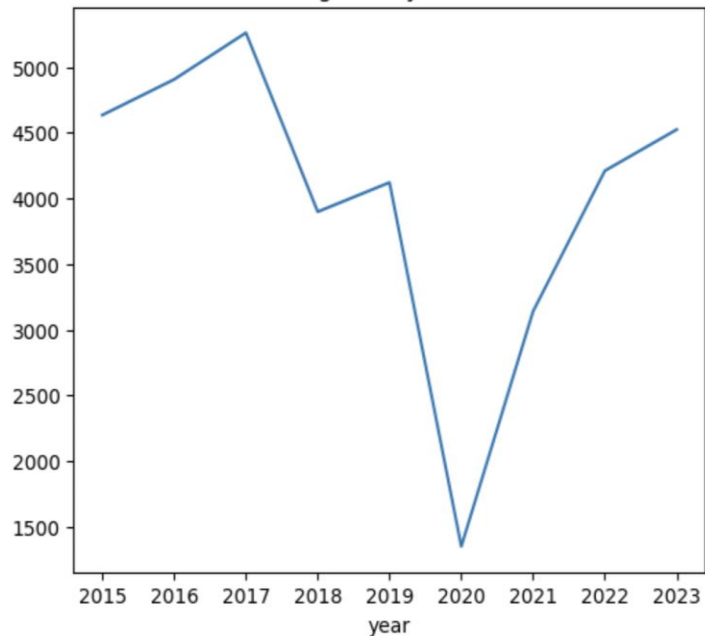


Total No. of Delay vs Carrier

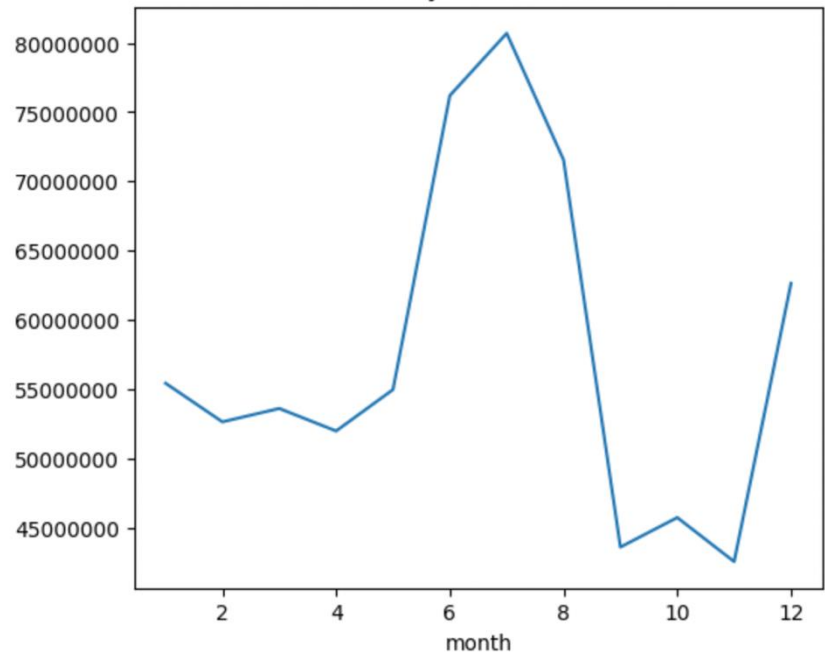


TRENDS

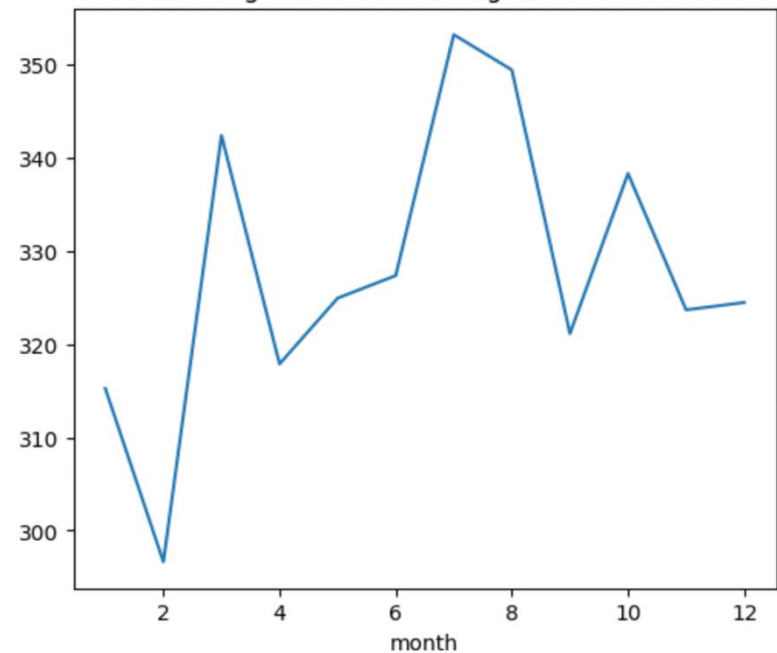
Average Delay vs Years



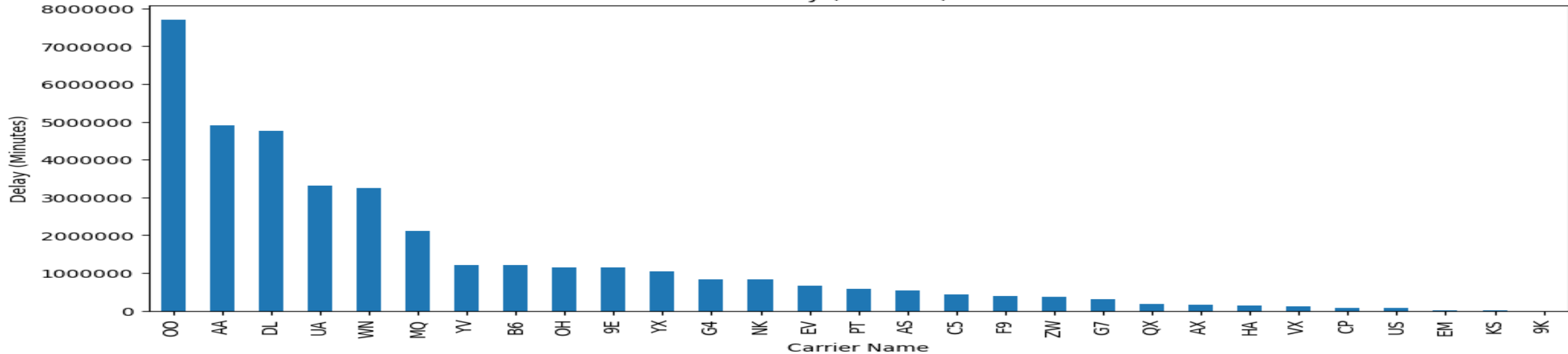
Total Delay Trend vs Months



Average No. of Arrival Flights vs Months



Weather Delay (Minutes) vs Carrier

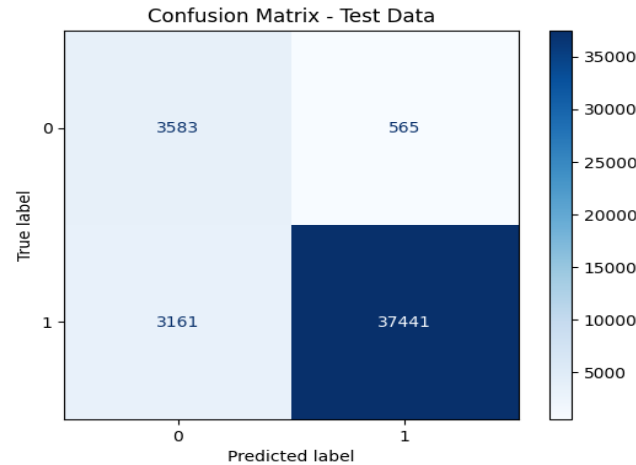


CLASSIFICATION PROBLEM

RANDOM FOREST CLASSIFIER IS USED FOR THE PREDICTION OF CLASSIFICATION PROBLEM

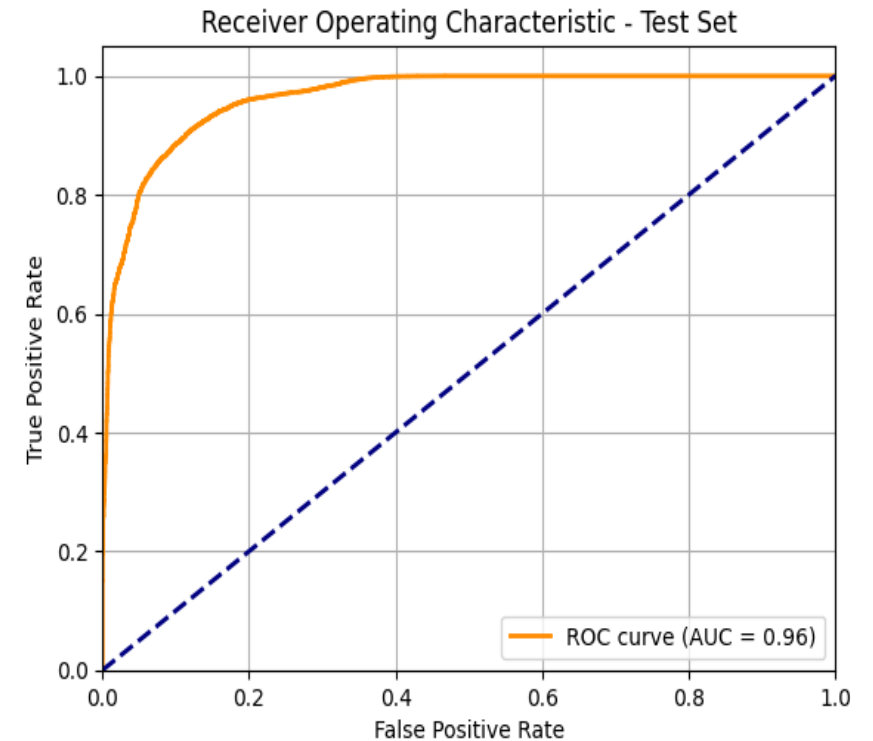
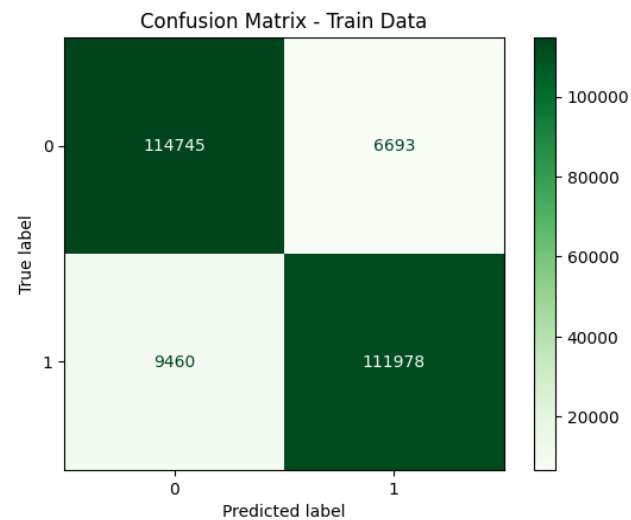
	precision	recall	f1-score	support
0	0.53	0.86	0.66	4148
1	0.99	0.92	0.95	40602
accuracy			0.92	44750
macro avg	0.76	0.89	0.81	44750
weighted avg	0.94	0.92	0.93	44750

TEST DATA



	precision	recall	f1-score	support
0	0.92	0.94	0.93	121438
1	0.94	0.92	0.93	121438
accuracy			0.93	242876
macro avg	0.93	0.93	0.93	242876
weighted avg	0.93	0.93	0.93	242876

TRAINING DATA



REGRESSION PROBLEM

RANDOM FOREST REGRESSOR IS USED FOR THE PREDICTION OF REGRESSION PROBLEM

DATA CLEANING & EXPLORATION

- Imputed 0.50% missing values in the age column using the **numerical median**, assuming they were **Missing at Random (MAR)** based on consistent age distribution before and after.
- Categorical columns were imputed using the **most frequent category**.
- Applied **one-hot encoding** to categorical features after imputation.
- No duplicate rows were found in the dataset.

FEATURE ENG. & CONSTRUCTION

- **arr_flights** – Arrival flights
- **month_sin** – Month sine
- **month_cos** – Month cosine
- **carrier_ct** – Carrier delay
- **weather_ct** – Weather delay
- **nas_ct** – NAS delay
- **security_ct** – Security delay
- **late_aircraft_ct** – Aircraft delay
- **carrier** – Airline code
- **airport** – Airport code

TARGET COLUMN

Created a new feature **OAI (Operational Accuracy Index)** as a weighted combination of:

60% weight to carrier_ct_rate and **40%** weight to late_aircraft_ct_rate

$$\text{OAI} = (0.6 \times \text{carrier_ct_rate}) + (0.4 \times \text{late_aircraft_ct_rate})$$

EVALUATING THE RANDOM FOREST CLASSIFIER

- R2 Score Test: 0.9963
- R2 Score Train: 0.9994
- MAE on Test Data: 0.00094
- MSE on Test Data: 0.00000814

REGRESSION USING OPERATIONAL ADJUSTABILITY INDEX (OAI)

DATA CLEANING & EXPLORATION

- **Dropped 0.50% missing values** in the age column assuming they were **Missing at Random (MAR)** based on stable distribution before and after removal.
- **No duplicate rows** were found in the dataset.
- **Density plot of age** remained consistent before and after dropping missing values, supporting the MAR assumption.

TARGET COLUMN

- arr_delay represents the **total arrival delay in minutes** for a given flight group.
- It serves as the **target variable** to predict overall delay severity using delay causes and flight attributes.

FEATURE ENG. & CONSTRUCTION

- arr_flights – Arrival flights
- month_sin – Month sine
- month_cos – Month cosine
- carrier_ct – Carrier delay
- weather_ct – Weather delay
- nas_ct – NAS delay
- security_ct – Security delay
- late_aircraft_ct – Aircraft delay
- carrier – Airline code
- airport – Airport code
- arr_del15 – Delay count
- arr_del15_rate – Delay rate
- carrier_ct_rate – Carrier rate
- weather_ct_rate – Weather rate
- nas_ct_rate – NAS rate
- security_ct_rate – Security rate
- late_aircraft_ct_rate – Aircraft rate
- avg_delay_per_flight – Avg delay

EVALUATING THE RANDOM FOREST CLASSIFIER

- R2 Score Test: 0.9816
- R2 Score Train: 0.9864
- MAE On Test Data : 0.0132
- MSE On Test Data: 0.0015

PARAMETERS USED

- n_estimators=600
- max_depth=25
- max_features='sqrt'
- min_samples_leaf=2
- n_jobs=-1
- random_state=42

A series of thin, light-brown lines forming an abstract geometric pattern in the top-left corner of the slide. The lines intersect to create various triangular and polygonal shapes.

THANK YOU

VIKRANT SINGH

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CHE 4Y [B.TECH.]

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